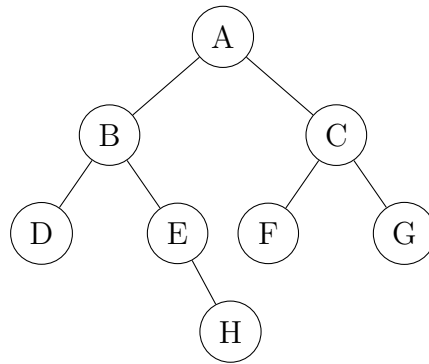


Quiz: Trees and Tree Traversals



Write the following traversals of this tree:

(a) Preorder traversal:

(b) Inorder traversal:

(c) Postorder traversal:

Question 2: Height of a Binary Tree

The **height** of a binary tree is defined as the length of the longest path from the root to any leaf, measured in number of edges.

Formally:

$$\text{height}(T) = \begin{cases} -1, & \text{if } T \text{ is empty} \\ 1 + \max(\text{height}(T.\text{left}), \text{height}(T.\text{right})), & \text{otherwise} \end{cases}$$

Given the following struct definition in C:

```
struct Node {
    int value;
    struct Node *left;
    struct Node *right;
};
```

Write a function `int height(struct Node *root)` that returns the height of the tree using recursion. Your function must correctly handle the case where `root == NULL`.